Geometrik shakllardagi noma'lum burchaklar

Bir chizmada uchburchak ham, to'rtburchak ham bo'lishi mumkin. Noma'lum burchakni bilgan qoidalarimiz bilan topamiz.

1-QISM

Mavzu

CHIZMANI QANDAY O'QIYMIZ

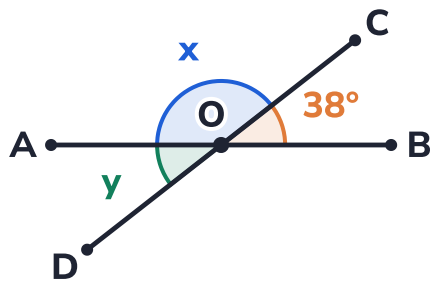
- $\angle ABC$ — uchi **B** da bo'lgan burchak.
- Bir xil chiziqchalar — teng tomonlar, strelkalar — parallel tomonlar.
- Tomonga yopishgan burchaklar — tomonning ikki uchidagi burchaklar.

1-QOIDA · CHIZIQ VA NUQTADAGI BURCHAKLAR

- To'g'ri chiziqdagi burchaklar yig'indisi 180° ga teng.
- Vertikal burchaklar teng.
- Nuqta atrofidagi burchaklar yig'indisi 360° ga teng.

1-MISOL · KESISHGAN TO'G'RI CHIZIQLAR

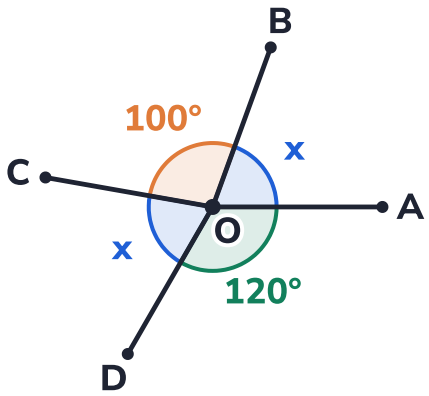
AB va CD to'g'ri chiziqlar O nuqtada kesishadi. $\angle BOC = 38^\circ$. x va y ni toping.



1. x va 38° — AB to'g'ri chiziqda: $x = 180^\circ - 38^\circ = 142^\circ$.
2. y va 38° — vertikal burchaklar: $y = 38^\circ$.

2-MISOL · NUQTA ATROFIDAGI BURCHAKLAR

O nuqta atrofida to'rtta burchak bor: x, 100°, x va 120°. x ni toping.



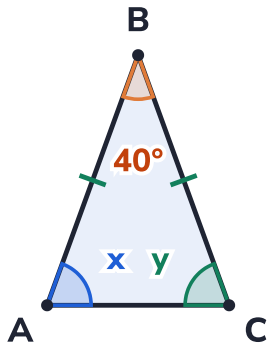
1. Burchaklar yig'indisi 360°: $x + x = 360^\circ - 100^\circ - 120^\circ = 140^\circ$.
2. $x = 140^\circ : 2 = 70^\circ$.

2-QOIDA · UCHBURCHAKNING BURCHAKLARI

- Uchburchakning ichki burchaklari yig'indisi 180° ga teng.
- Teng tomonli uchburchakning 3 ta teng tomoni va 3 ta teng burchagi bor. Har bir burchagi 60° ga teng.
- Teng yonli uchburchakning 2 ta teng tomoni va 2 ta teng burchagi bor. Uning teng tomonlari qarshisidagi burchaklari teng.
- To'g'ri burchakli uchburchak bitta to'g'ri burchakka ega. Uning qolgan 2 burchagining yig'indisi 90° ga teng.

1-MISOL · TENG YONLI UCHBURCHAK

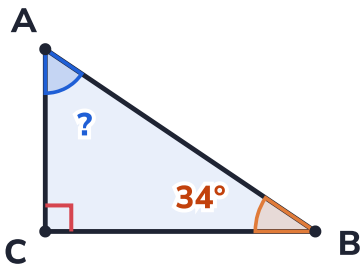
ABC — teng yonli uchburchak, AB = BC, $\angle B = 40^\circ$. x va y ni toping.



1. Teng yonli uchburchakda x = y.
2. $x + y = 180^\circ - 40^\circ = 140^\circ$.
3. $x = y = 140^\circ : 2 = 70^\circ$.

2-MISOL · TO'G'RI BURCHAKLI UCHBURCHAK

ABC uchburchakda $\angle C = 90^\circ$, $\angle B = 34^\circ$. $\angle A$ ni toping.



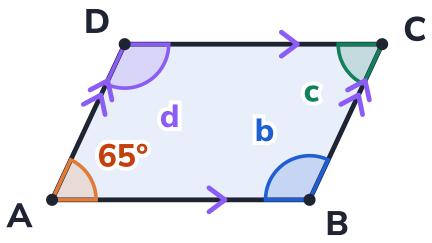
1. Qolgan ikki burchak yig'indisi 90°: $\angle A + \angle B = 90^\circ$.
2. $\angle A = 90^\circ - 34^\circ = 56^\circ$.

3-QOIDA · TO'RTBURCHAKNING BURCHAKLARI

- Kvadrat va to'g'ri to'rtburchakning har bir burchagi 90° ga teng.
- Parallelogrammning (rombning): a) ixtiyoriy tomoniga yopishgan burchaklari yig'indisi 180° ga teng; b) qarama-qarshi burchaklari teng.
- Trapetsiyaning har bir yon tomoniga yopishgan burchaklari yig'indisi 180° ga teng.

1-MISOL · PARALLELOGRAMM

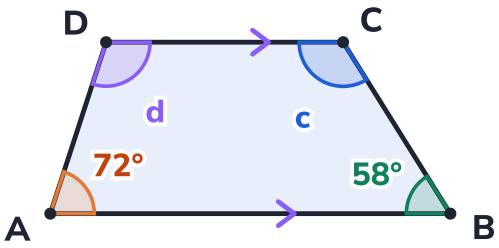
ABCD — parallelogramm, $\angle A = 65^\circ$. b, c va d burchaklarni toping.



1. AB tomoniga yopishgan burchaklar: $b = 180^\circ - 65^\circ = 115^\circ$.
2. Qarama-qarshi burchaklar teng: $c = 65^\circ$, $d = b = 115^\circ$.

2-MISOL · TRAPETSIYA

ABCD — trapetsiya, AB va DC — asoslari. $\angle A = 72^\circ$, $\angle B = 58^\circ$. c va d ni toping.



1. AD yon tomoniga yopishgan burchaklar: $d = 180^\circ - 72^\circ = 108^\circ$.
2. BC yon tomoniga yopishgan burchaklar: $c = 180^\circ - 58^\circ = 122^\circ$.

Chizmada shakllar ko'p bo'lsa, qaysi burchakdan boshlayman?

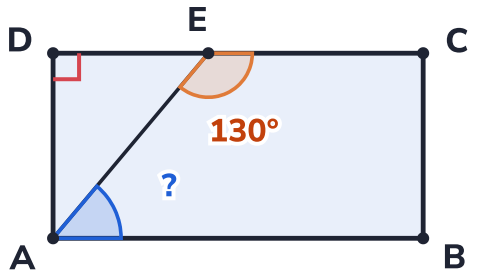


ISH TARTIBI

Ma'lum burchakdan boshlang. Har qadamda bitta burchak toping — izlanganiga yetguncha.

1-MISOL · TO'G'RI TO'RTBURCHAK VA KESMA

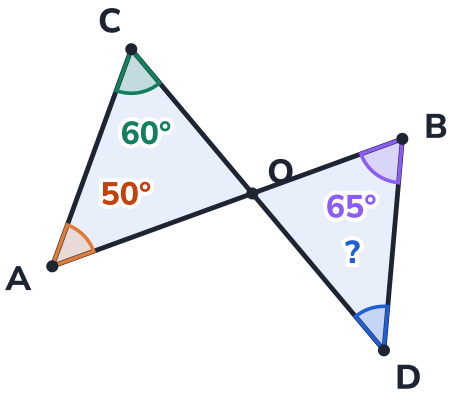
ABCD — to'g'ri to'rtburchak. E nuqta DC tomonda yotadi, $\angle AEC = 130^\circ$. $\angle EAB$ ni toping.



1. DC — to'g'ri chiziq: $\angle AED = 180^\circ - 130^\circ = 50^\circ$.
2. ADE uchburchakda $\angle D = 90^\circ$: $\angle DAE = 180^\circ - 90^\circ - 50^\circ = 40^\circ$.
3. $\angle A = 90^\circ$ (to'g'ri to'rtburchak): $\angle EAB = 90^\circ - 40^\circ = 50^\circ$.

2-MISOL · IKKI UCHBURCHAK VA VERTIKAL BURCHAKLAR

AB va CD to'g'ri chiziqlar O nuqtada kesishadi. $\angle A = 50^\circ$, $\angle C = 60^\circ$, $\angle B = 65^\circ$. $\angle D$ ni toping.



1. AOC uchburchakda: $\angle AOC = 180^\circ - 50^\circ - 60^\circ = 70^\circ$.
2. Vertikal burchaklar teng: $\angle BOD = 70^\circ$.
3. BOD uchburchakda: $\angle D = 180^\circ - 65^\circ - 70^\circ = 45^\circ$.

2-QISM

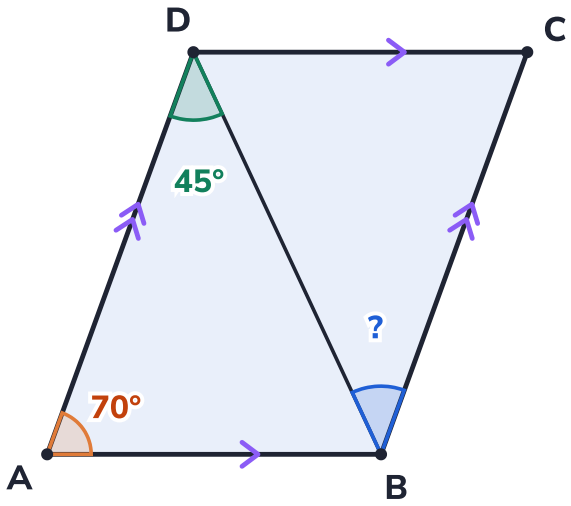
Misollar ishlash



Topgan burchagimni darrov chizmaga yozib qo'yaman — keyingi qadamda kerak bo'ladi.

1-MISOL · PARALLELOGRAMM VA DIAGONAL

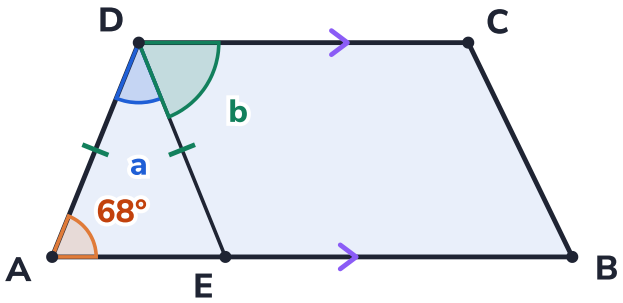
ABCD — parallelogramm, BD — uning diagonali. $\angle A = 70^\circ$, $\angle ADB = 45^\circ$. $\angle DBC$ ni toping.



1. ABD uchburchakda: $\angle ABD = 180^\circ - 70^\circ - 45^\circ = 65^\circ$. *(Uchburchak burchaklari yig'indisi 180°.)*
2. $\angle ABC = 180^\circ - 70^\circ = 110^\circ$. *(Parallelogrammda AB tomoniga yopishgan burchaklar yig'indisi 180°.)*
3. $\angle DBC = 110^\circ - 65^\circ = 45^\circ$.

2-MISOL · TRAPETSIYA ICHIDA TENG YONLI UCHBURCHAK

ABCD — trapetsiya, AB va DC — uning asoslari. E nuqta AB asosda yotadi va AD = DE. $\angle A = 68^\circ$. $a = \angle ADE$ va $b = \angle EDC$ ni toping.

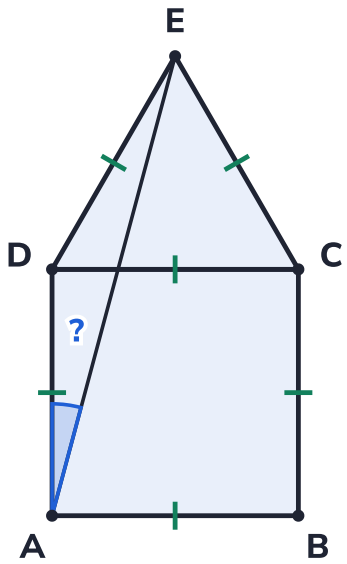


1. AD = DE, demak qarshisidagi burchaklar teng: $\angle DEA = \angle A = 68^\circ$.

2. $a = \angle ADE = 180^\circ - 68^\circ - 68^\circ = 44^\circ$.
3. $\angle ADC = 180^\circ - 68^\circ = 112^\circ$. *(Trapetsiyaning AD yon tomoniga yopishgan burchaklar.)*
4. $b = \angle EDC = 112^\circ - 44^\circ = 68^\circ$.

3-MISOL · KVADRAT VA TENG TOMONLI UCHBURCHAK

ABCD — kvadrat, DCE — teng tomonli uchburchak. A va E nuqtalar tutashtirilgan. $\angle DAE$ ni toping.



1. D nuqtadagi burchak ikki bo'lakdan iborat: $\angle ADE = 90^\circ + 60^\circ = 150^\circ$. *(Kvadrat burchagi va teng tomonli uchburchak burchagi.)*
2. AD = DC (kvadrat tomonlari), DC = DE (teng tomonli uchburchak tomonlari). Demak, AD = DE: ADE teng yonli.
3. $\angle DAE = \angle DEA = (180^\circ - 150^\circ) : 2 = 15^\circ$.



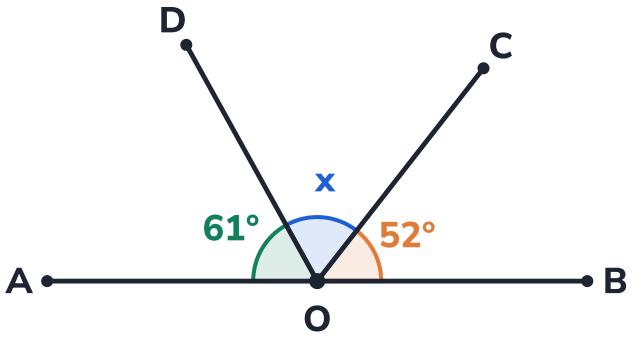
DIQQAT!

1. Teng yonli uchburchakda teng burchaklar — teng tomonlar **qarshisida** turadi. Teng tomonlar orasidagi burchak boshqacha bo'lishi mumkin.
2. Qarama-qarshi burchaklar parallelogrammda (rombda, to'g'ri to'rtburchakda, kvadratda) teng. Trapetsiyada qarama-qarshi burchaklarni teng deb olmang — u parallelogramm emas.
3. Burchakni ko'z bilan chamalamang — har bir sonni hisoblab toping.

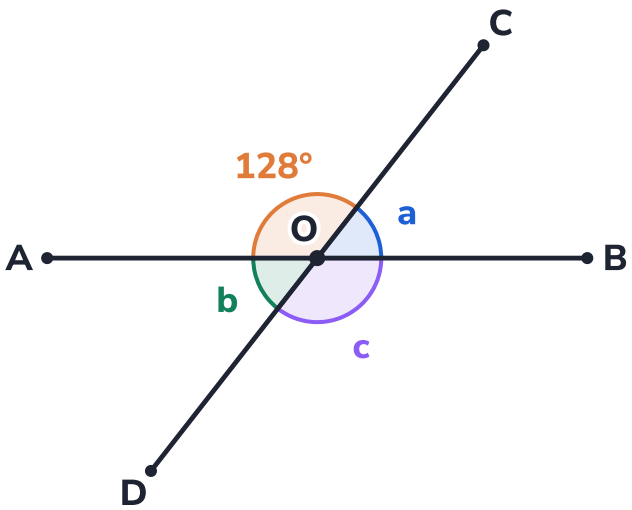
3-QISM

Mustahkamlash

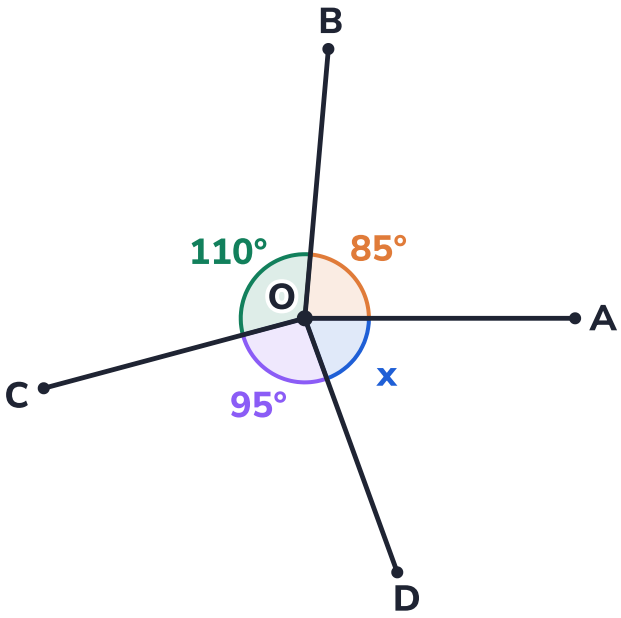
1. AB — to'g'ri chiziq. x ni toping.



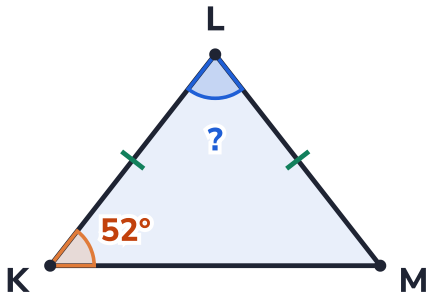
2. AB va CD to'g'ri chiziqlar O nuqtada kesishadi, $\angle AOC = 128^\circ$. a, b va c ni toping.



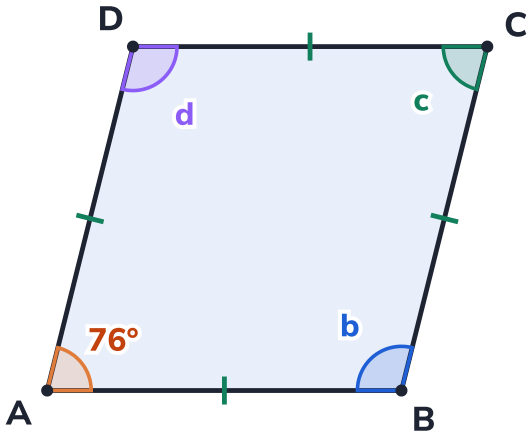
3. O nuqta atrofidagi x burchakni toping.



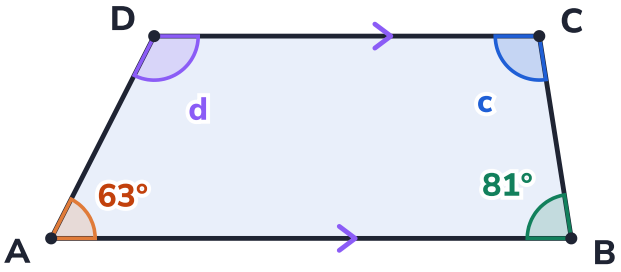
4. KLM — teng yonli uchburchak, KL = LM, $\angle K = 52^\circ$. $\angle L$ ni toping.



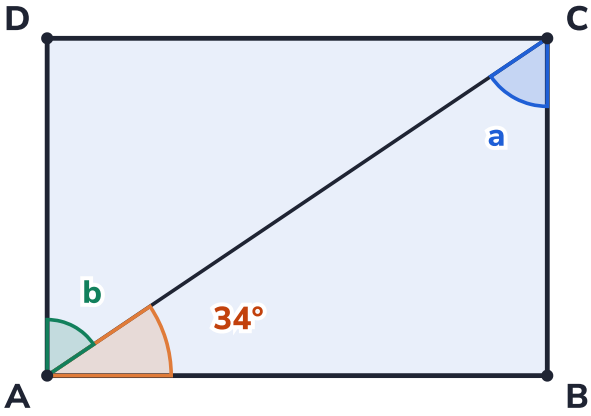
5. ABCD — romb, $\angle A = 76^\circ$. b, c va d ni toping.



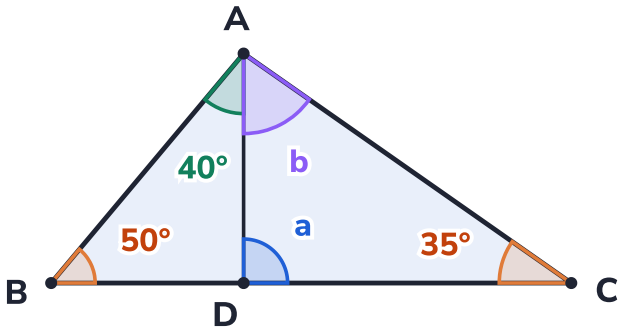
6. ABCD — trapetsiya, AB va DC — asoslari. $\angle A = 63^\circ$, $\angle B = 81^\circ$. c va d ni toping.



7. ABCD — to'g'ri to'rtburchak, AC — diagonal, $\angle BAC = 34^\circ$. $a = \angle ACB$ va $b = \angle CAD$ ni toping.



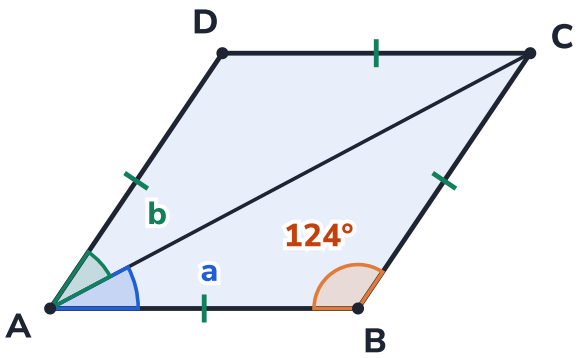
8. ABC uchburchakda D nuqta BC tomonda yotadi. $\angle B = 50^\circ$, $\angle BAD = 40^\circ$, $\angle C = 35^\circ$. $a = \angle ADC$ va $b = \angle DAC$ ni toping.



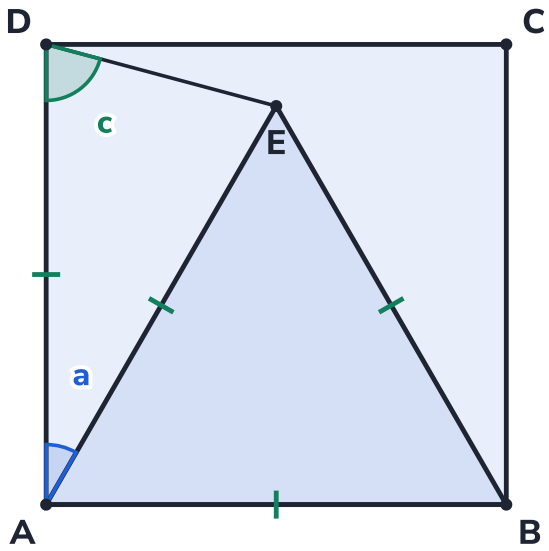
4-QISM

Mustaqil ish

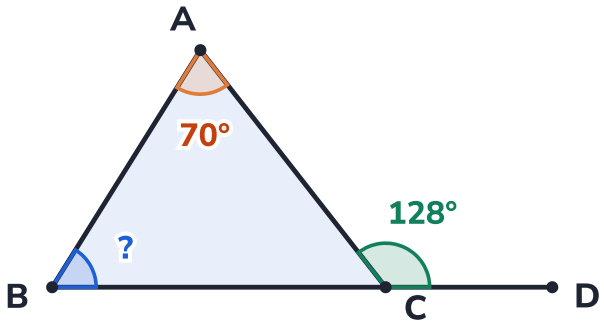
1. ABCD — romb, AC — diagonal, $\angle ABC = 124^\circ$. $a = \angle BAC$ va $b = \angle CAD$ ni toping.



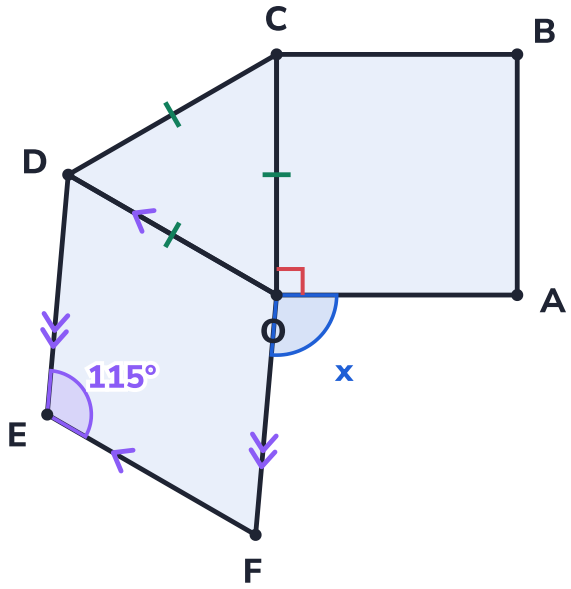
2. ABCD — kvadrat, ABE — kvadrat ichidagi teng tomonli uchburchak. $a = \angle EAD$, $c = \angle ADE$ va $\angle EDC$ ni toping. *(Maslahat: AB, AD va AE tomonlarni solishtiring.)*



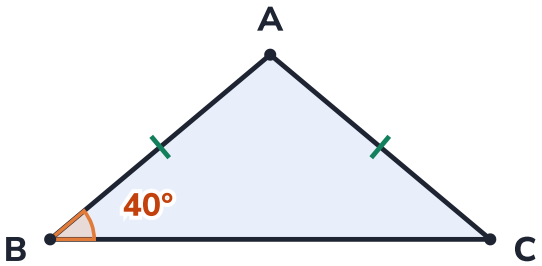
3. ABC uchburchakning BC tomoni D nuqtagacha davom ettirilgan. $\angle A = 70^\circ$, $\angle ACD = 128^\circ$. $\angle B$ ni toping.



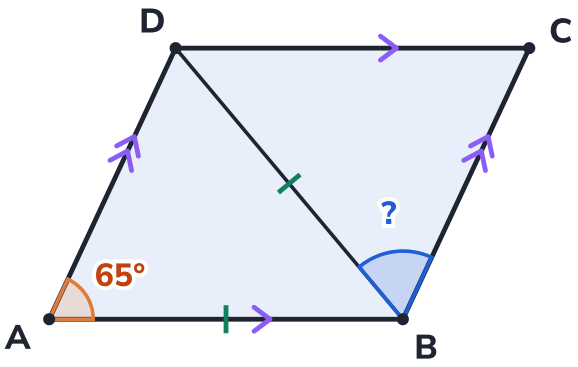
4. O nuqtada kvadrat OABC, teng tomonli uchburchak OCD va parallelogramm ODEF uchrashadi. $\angle E = 115^\circ$. x ni toping. *(Maslahat: O nuqta atrofida 4 ta burchak bor.)*



5. **Xatoni toping.** ABC — teng yonli uchburchak, AB = AC, $\angle B = 40^\circ$. Sardor shunday yozdi: «$\angle A = 40^\circ$, chunki teng yonli; $\angle C = 180^\circ - 40^\circ - 40^\circ = 100^\circ$». Sardorning xatosi nimada? To'g'ri javobni toping.



6. **Qiyinroq.** ABCD — parallelogramm, BD — diagonal, AB = BD. $\angle A = 65^\circ$. $\angle DBC$ ni toping.



5-QISM

Javoblar

Mustahkamlash

1-masala

$x = 180^\circ - 52^\circ - 61^\circ = 67^\circ$.

2-masala

$a = 180^\circ - 128^\circ = 52^\circ$ (to'g'ri chiziq); $b = a = 52^\circ$, $c = 128^\circ$ (vertikal burchaklar).

3-masala

$x = 360^\circ - 85^\circ - 110^\circ - 95^\circ = 70^\circ$.

4-masala

$\angle M = \angle K = 52^\circ$; $\angle L = 180^\circ - 52^\circ - 52^\circ = 76^\circ$.

5-masala

$b = 180^\circ - 76^\circ = 104^\circ$; $c = 76^\circ$, $d = 104^\circ$ (qarama-qarshi burchaklar).

6-masala

$d = 180^\circ - 63^\circ = 117^\circ$ (AD yon tomon); $c = 180^\circ - 81^\circ = 99^\circ$ (BC yon tomon).

7-masala

ABC uchburchakda $\angle B = 90^\circ$: $a = 180^\circ - 90^\circ - 34^\circ = 56^\circ$. $\angle A = 90^\circ$: $b = 90^\circ - 34^\circ = 56^\circ$.

8-masala

ABD uchburchakda: $\angle ADB = 180^\circ - 50^\circ - 40^\circ = 90^\circ$. BC — to'g'ri chiziq: $a = 180^\circ - 90^\circ = 90^\circ$. ADC uchburchakda: $b = 180^\circ - 90^\circ - 35^\circ = 55^\circ$.

Mustaqil ish

1-masala

AB = BC (romb tomonlari) — ABC teng yonli: $a = (180^\circ - 124^\circ) : 2 = 28^\circ$. $\angle BAD = 180^\circ - 124^\circ = 56^\circ$, $b = 56^\circ - 28^\circ = 28^\circ$.

2-masala

$\angle EAB = 60^\circ$, $a = 90^\circ - 60^\circ = 30^\circ$. AD = AB = AE — ADE teng yonli: $c = (180^\circ - 30^\circ) : 2 = 75^\circ$. $\angle EDC = 90^\circ - 75^\circ = 15^\circ$.

3-masala

$\angle ACB = 180^\circ - 128^\circ = 52^\circ$ (to'g'ri chiziq); $\angle B = 180^\circ - 70^\circ - 52^\circ = 58^\circ$.

4-masala

$\angle DOF = \angle E = 115^\circ$ (parallelogrammning qarama-qarshi burchaklari). $x = 360^\circ - 90^\circ - 60^\circ - 115^\circ = 95^\circ$.

5-masala

Teng tomonlar AB va AC; ularning qarshisidagi burchaklar — B va C. Demak, $\angle C = \angle B = 40^\circ$, $\angle A = 180^\circ - 40^\circ - 40^\circ = 100^\circ$. Sardor A burchakni teng deb oldi — u teng tomonlar orasida turadi.

6-masala

ABD teng yonli (AB = BD): $\angle ADB = \angle A = 65^\circ$; $\angle ABD = 180^\circ - 65^\circ - 65^\circ = 50^\circ$. $\angle ABC = 180^\circ - 65^\circ = 115^\circ$; $\angle DBC = 115^\circ - 50^\circ = 65^\circ$.